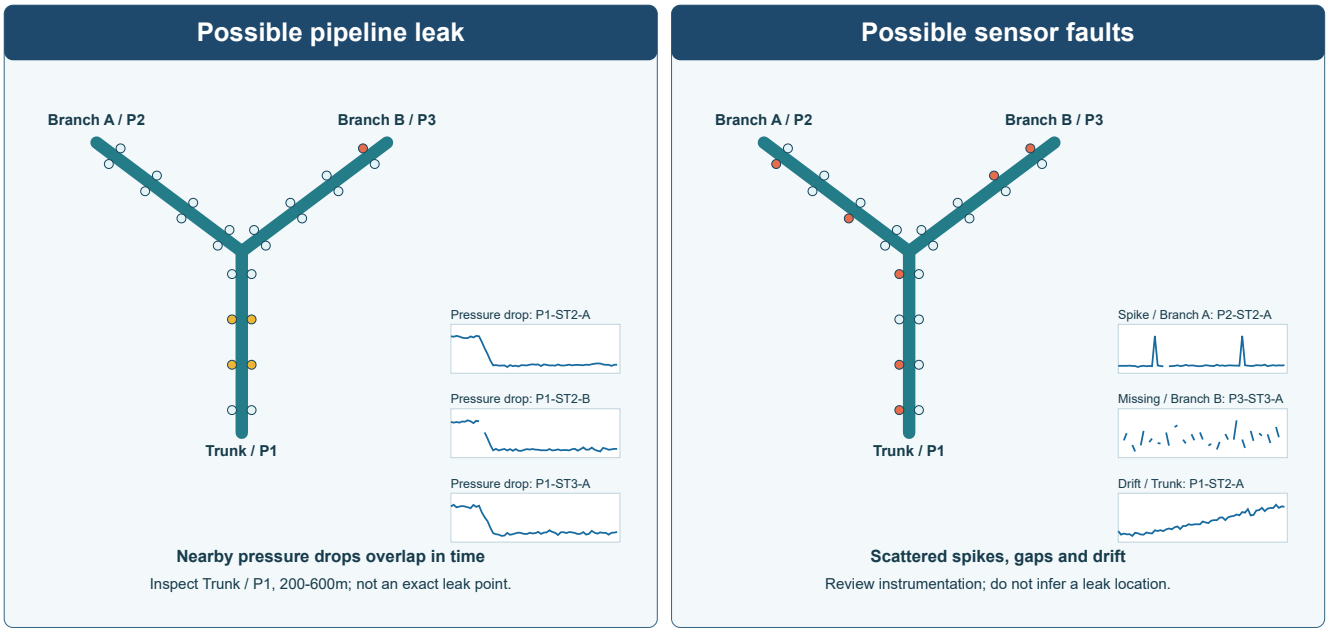
Subsea Integrity Copilot - Customer Brief

Prospect and problem. NorthSea Integrity Operations is a fictional offshore gas operator. Engineers must distinguish pipeline leaks from sensor spikes, gaps and drift. Missed leaks threaten the environment and operations; false alarms trigger costly inspections. Scarce expertise is difficult to encode and transfer to newcomers.



Existing detector alerts → Agent-assisted investigation → Human review

Patterns are evidence, not proof: check operating changes and shared failures.

Demo assumptions: 24 pipe-mounted pressure instruments, 12 paired stations, three connected 800m segments.

Same pressure-pack-v3 geometry as console; seed 2026, [80,140)s, pressure in bar; independent trace y-scales; synthetic.

Why this problem. My PhD studied marine data quality, anomaly detection and translating expertise into smart systems. This applies that experience to reviewable investigation.

The two customer roles. The **Head of Subsea Operations** owns the budget: cost, workload, timeline and risk. The **Lead Subsea Integrity & Instrumentation Engineer** challenges calibration, shared failures, missing data and unsupported conclusions. Integrity/instrumentation engineers and newcomers use the copilot.

Why unsigned. Doubts about effort savings, hallucinations, missed leaks, integration cost, privacy and accountability. Cheaper fixed workflows are credible alternatives.

Working demo. An upright Y has three 800m pipes and 24 pressure instruments in 12 paired stations. Frozen median/MAD detection feeds sliding-window timing, topology and failure-domain analysis. Nearby sustained drops support inspection intervals; scattered faults trigger sensor review; inadequate evidence withholds conclusions. Questions: "Leak or faulty sensors?", "Where should we inspect, and why?" and "What evidence supports this, and what remains unknown?"

Agent value. Three scoped roles investigate quality, choose incident-time history searches/comparisons and explain competing causes. A separate same-model critic reviews original sources; code checks quotes, coverage and result identity. Revise once, then accept or withhold. Engineers inspect sources, ask follow-ups and download evidence. Reviewed procedures support knowledge reuse, not automatic expertise acquisition.

Limits and buying gate. Fictional dirty/conflicting data; unvalidated hydraulics; intervals, not exact leak positions. Hosted reports: 15/18 first attempts passed, then 3/3 explicit retries. Fixed reports: 18/18 and faster. No field-accuracy or hallucination-free claim. Numerical warnings never wait for inference. Assume ten-minute manual analysis; target five-minute reports, counting failures and human review separately. Propose a four-week read-only shadow pilot with agreed accuracy, abstention, unsupported-claim, latency and cost gates. Pricing requires discovery; savings are unproven and connectors unbuilt. Hosted inference requires approval. Humans retain safety authority. Buy only if quality, effort and total cost beat manual/fixed baselines.

Private GitHub and run instructions: github.com/ntnguyen-so/subsea-integrity-copilot. Repository access must be granted before review.